

Phases of the Moon

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus The eight moon phases repeat in order as the Moon orbits Earth; waxing means the illuminated portion we see is growing, while waning means it is shrinking.

Round 1 - Retrieve It

1. What does waxing mean?

2. What does waning mean?

3. What phase comes after new moon?

4. What phase comes after full moon?

Round 2 - Sketch It

Make an eight-position phase sequence and label: new, waxing crescent, first quarter, waxing gibbous, full, waning gibbous, third quarter, waning crescent.

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Phases of the Moon - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which phase comes between waxing crescent and waxing gibbous?

2. Which phase comes between waning gibbous and waning crescent?

3. Does the Moon produce its own visible light?

Round 4 - Reason from Evidence

Explain why ordinary moon phases are not caused by Earth's shadow falling on the Moon.

Phases of the Moon - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Phases of the Moon in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect phases to the Moon's orbit. Why does the sequence repeat?

E**Extend**

If the Moon is at first quarter tonight, predict the general trend in visible illumination over the next several days.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Phases of the Moon

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The visible illuminated portion is increasing.
2. The visible illuminated portion is decreasing.
3. Waxing crescent.
4. Waning gibbous.

Sketch It - look for

Correct cyclic order. Exact shading orientation depends on viewing convention/hemisphere; phase names and waxing/waning sequence matter most.

Apply It

1. First quarter.
2. Third quarter.
3. No. We see sunlight reflected from the Moon.

Reason from Evidence - look for

Phases result from changing viewing geometry of the Moon's sunlit half as it orbits Earth. Earth's shadow causes a lunar eclipse, not the regular monthly phases.

ACE - Extend look-for

It will continue waxing toward waxing gibbous and then full moon.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.